

DSA5102 Final Group Project

Due: November 22nd

1 Preface

The goal of the course project is to familiarize yourself with

1. Basic Python data manipulation and visualization tools, such as NumPy, pandas, Matplotlib and seaborn.
2. Popular machine learning libraries such as scikit-learn and Keras.
3. Scientific methodologies in data analysis and machine learning.

2 Instructions

2.1 Project Submission Format

Your project should include **one Jupyter notebook** (also referred to as “project report” thereafter) and one **video presentation** for the entire group. Submit both parts on Canvas. For the Jupyter notebook, take note of the following:

1. Your submitted notebook should be runnable as it is. If some parts require long training, then please attach a saved model that can be loaded into the notebook and write in your code *TRAIN_MODEL = False*.
2. Include clear comments (use the markdown capabilities of jupyter!) to explain what you are doing at each step, and also your findings and how they may be interesting. Treat it as an “essay”!
3. Complete the project in groups of 4-5 people. **Individual submission is not allowed.** Each group should submit only one report together. Within the report, declare how different members contributed to the project. There are two possibilities:
 - If you declare “all members contributed equally” (or if you don’t make any declaration at all), then everyone in the group will get a similar grade for the project report.
 - Alternatively, if you write “xxx contributed to y...”, then group members will get different scores depending on your individual contribution.
4. The use of Generative Artificial Intelligence must also be clearly indicated within the report. This includes a brief description of how and for what purposes you use the AI. **If you declare that generative AI is not used, but we find reliable evidence on your AI usage, this will be considered plagiarism and your score for both parts of the final project will be zero.**
5. Upload your dataset alongside your Jupyter notebook and video presentation if possible. If your dataset is too large, include a link of your dataset in your Jupyter notebook.

For the video presentation, the requirements are as follows:

1. The total time of the video presentation is 20 minutes for a group of 5 people, and 16 minutes for a group of 4 people. Every person must present for roughly 4 minutes. Each group should submit a single video that contains the presentation of every group member.
2. The presentation should be based on your group project, which should include the motivation and visualization of the dataset, as well as the methodology, algorithms and implementation results. There is no need to run any part of the code during your presentation.
3. The purpose of the presentation is for us to see how much you understand your project report (which may be largely AI-generated). If you do not seem familiar with your part of the report, this will negatively impact your score in both project report and presentation parts.
4. The presentation must be done by real human. Synthetic presenters or voices are prohibited. **If we find credible evidence that your presentation is generated by any computer tool, your score for both parts of the final project will be zero.**
5. Slides are unnecessary, but you can make some slides if you believe this will help your presentation. There's no need to submit your slides.

2.2 The Task

There are two project options, and you should complete one of the following:

1. Choose a data set and complete three different data analysis tasks in supervised and unsupervised learning, with at least one supervised learning task (e.g. regression, classification, etc), and one unsupervised learning task (e.g. dimensionality reduction, clustering, generative models, etc). The third task can be on either supervised or unsupervised learning.
 - For each task, employ at least two different methods and compare their performance. For supervised learning task(s), cross-validation should be used for model selection. For unsupervised learning tasks, use internal metrics (for example, Silhouette scores or within-cluster MSE for clustering), visualization or project-based interpretation. Different choices of hyperparameters for the same algorithm do not count as different methods.
 - If your report contains two (un)supervised learning tasks, then at least 3 different (un)supervised learning methods must be used within the entire project.
 2. Train the machine to play some classic/well-known game using reinforcement learning. Some examples are Breakout, Snake, Tetris and Super Mario. Some of the games can be found in Atari Games.
 - You should demonstrate visually how the computer's gameplay improves as the training progresses.
 - Explore how you could improve the strength of the computer player, and compare their performances.
 - Include a baseline, learning curves, and evaluation over multiple episodes in your report.
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For both options, you should include an introduction of your dataset/game (if you choose option 2, assume that I have never played the game before!), with visualizations.

You are also welcome to explore alternative options for the final project. Email me if you have some creative ideas, but do not directly fit into the previous two options, and are not sure if this would be accepted.

3 Grading Criteria

The total grade is 20 points for project report and 10 points for video presentation. Your grade for the project report is based on the following:

1. Completion of the aforementioned tasks.
2. Scientific correctness of your methodologies. For reinforcement learning, experimental rigor.
3. Correct use of libraries.
4. Clear documentation of introduction/visualization, code and findings. If you choose option 2, the complexity of the game and the strength of your final output are important.
5. Creativity and style.

For the video presentation, your grade is based on the following:

1. Clarity and coherence.
2. Understanding of your project.
3. Time usage. If the group is 2 minutes or more overtime, then the score of each group member will be deducted, depending on the actual time usage of each individual member. In principle, 0.5 points is deducted for each member of a group for every 30 seconds being overtime.
4. Your score for presentation and project report may be correlated.

4 Datasets

You are encouraged to find a dataset with an application area that interests you, as long as we never used this dataset in class demos. You may browse the following resources:

1. UCI Machine Learning Repository
2. Kaggle
3. data.gov.sg

In case you have difficulty finding a dataset, I have included some possible choices below, but don't limit yourself to these options!

1. Heart Disease
 2. Retail Data Analytics
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3. Atomization Energies and DFT
 4. New York Stock Exchange
 5. Traffic Signs
 6. COVID-19
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